

Secure Exit

SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

PENUGONDA JEEVAN KUMAR
192472082RD

CO3-AT3-Assignment

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QUESTIONS

Q1

Banking Transaction System –
Exception Handling
10 marks

Q2

Hospital Patient Registration –
User-Defined Exception
10 marks

Q3

E-Commerce Order Processing
– Built-in Exceptions
10 marks

Q4

Hospital Emergency System –
Thread Priority
10 marks

Q5

Online Food Delivery – Multiple
Threads and Execution Order
10 marks

5/5 answered

Q1

Banking Transaction System –
Exception Handling10
marks

Develop a Java program for an ATM withdrawal system. The program accepts the account balance and withdrawal amount.

The program must:

- Accept the current account balance and withdrawal amount.
- Throw a user-defined `InsufficientBalanceException` when the withdrawal amount exceeds the balance.
- Handle withdrawal amounts that are zero or negative.
- Handle invalid numeric input using a suitable built-in exception.
- Display a meaningful transaction result.
- Use try-catch-finally so that the transaction completion message is always displayed.

Sample Test Cases:

Input / Scenario	Expected Result	Purpose
10000 3000	Normal case	
10000 12000	Boundary / exception case	
10000 -500	Invalid input	
abc 3000	Invalid input	

Your Code

```
1 class Main{
```

```
2    public static void main(String[] args)
3        double balance = 10000, amount = 3000;
4        try{
5            if(amount <= 0) throw new IllegalArgumentException("Amount must be greater than 0");
6            if(amount > balance) throw new InsufficientBalanceException("Amount must be less than balance");
7        } catch (Exception e){
8            System.out.println(e.getMessage());
9        } finally {
10            System.out.println("Transaction completed");
11        }
12    }
13 }
14 class InsufficientBalanceException extends IllegalArgumentException{
15     InsufficientBalanceException(String s){
16         super(s);
17     }
18 }
```



Input

10000
3000



Output

Transaction completed

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5/5 answered

Q2

Hospital Patient Registration – User-Defined Exception

10
marks

Develop a Java program for a hospital patient registration system. The program accepts the patient's name, age, and body temperature. Create suitable user-defined exception classes for invalid age and invalid temperature. Validate the input before registration. Use try-catch-finally and display a registration completion message from finally.

Sample Test Cases:

Input / Scenario Expected Result Purpose

Arun, 35, 98.6 Patient registered successfully Normal case

Meena, 12, 98.4 InvalidAgeException Boundary / exception case

Ravi, 45, 105.2 InvalidTemperatureException Invalid input

Kumar, -2, 98.1 InvalidAgeException Invalid input

Your Code

```
1 class Main{
2     public static void main(String[] args)
3         int age = 35;
4         double temp = 98.6;
5         try{
6             if(age < 18) throw new Invalid
```

```
7         if(temp < 95 || temp > 100) th
8             System.out.println("Patient re
9         } catch (Exception e){
10             System.out.println(e.getMessag
11         } finally {
12             System.out.println("Registatio
13         }
14
15     }
16 }
17 class InvalidAgeException extends Exceptio
18     InvalidAgeException() {
19         super("InvalidAgeException");
20     }
21 }
22 class InvalidTemperatureException extends
23     InvalidTemperatureException() {
24         super("InvalidTemperatureExceptio
25     }
26 }
```



Input

Arun 35 98.6



Output

Patient registerd successfully
Registation completed

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Threads and Execution Order

10 marks



5/5 answered

Q3

E-Commerce Order Processing – Built-in Exceptions

10
marks

Develop a Java program for an online shopping application. Store product names and prices in arrays and accept a product index and quantity from the customer.

The program must:

- Calculate total cost as quantity × price.
- Handle invalid product indices.
- Handle invalid numeric input.
- Reject zero or negative quantities.
- Use appropriate built-in exception classes.
- Continue execution after handling an invalid transaction.

Sample Test Cases:

Input / Scenario Expected Result Purpose

1, 2 Valid product; total cost calculated Normal case

5, 2 ArrayIndexOutOfBoundsException handled

Boundary / exception case

2, -1 Invalid quantity Invalid input

abc, 2 InputMismatchException handled Invalid input

Your Code

```
1 class Main{
2     public static void main(String[] args)
3         int index = 1, qty = 2;
4         double[] price = {100, 200, 300};
```

```
5         try{
6             if(qty <= 0) throw new Illegal
7             System.out.println("Total cost
8         } catch (ArrayIndexOutOfBoundsException
9             System.out.println("Invalid p
10        } catch (IllegalArgumentException
11            System.out.println(e.getMessag
12        }
13    }
14 }
```



Input

```
1
2
100 200 300
```



Output

```
Total cost: 400.0
```

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10 marks

5/5 answered

Q4

**Hospital Emergency System – Thread
Priority**

10

marks

A hospital application creates threads for Emergency Patient Processing, General Patient Processing, and Billing Processing.

Develop a Java program that assigns different priorities to these threads. Run the program several times and analyze whether the emergency thread is guaranteed to execute first.

Then modify the design if the hospital requires a strict execution sequence rather than merely expressing scheduling preference.

Your answer must contain Java code and an execution analysis based on the actual thread behavior.

Sample Test Cases:

Input / Scenario Expected Result Analysis Focus
Emergency, General, Billing All threads execute;
priority does not guarantee a fixed output order
Normal execution

Emergency workload > General workload Emergency
thread receives a higher priority but exact ordering is
not guaranteed Concurrency / design
Strict emergency-before-billing requirement Use

explicit coordination rather than relying only on
priority Exception / boundary

Your Code

```
1 class Main{
2     static void order(){
3         System.out.println("Order Process:");
4     }
5     static void payment(){
6         System.out.println("Payment Proces");
7     }
8     static void delivery(){
9         System.out.println("Delivery Assig");
10    }
11    public static void main(String[] args){
12        Thread t1 = new Thread(() -> order);
13        Thread t2 = new Thread(() -> payment);
14        Thread t3 = new Thread(() -> delivery);
15        t1.start();
16        t1.join();
17        t2.start();
18        t2.join();
19        t3.start();
20        t3.join();
21    }
22 }
23 }
```

Input

stdin input

Output

Order Processing
Payment Processing
Delivery Assignment

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5/5 answered

Q5

Online Food Delivery – Multiple Threads
and Execution Order10
marks

An online food-delivery application uses three threads: Order Processing, Payment Processing, and Delivery Assignment.

Develop a Java program in which these activities are represented by separate threads. Analyze the problem if all three threads are started independently.

Modify the program so that:

- Order processing completes before payment processing.
- Payment processing completes before delivery assignment.
- The program uses appropriate Java thread-management techniques.
- The output clearly shows the required execution order.

Explain the difference between simply starting all threads and explicitly controlling their execution order.

Sample Test Cases:

Input / Scenario Expected Result Analysis Focus
Order=Pizza; Payment=UPI; Delivery=Agent A Order
→ Payment → Delivery Normal execution
Order=Burger; Payment=Card; Delivery=Agent B

Order → Payment → Delivery Concurrency / design
 Payment=Failed Delivery assignment should not
 proceed Exception / boundary

Your Code

```

1 class Main{
2     static String b;
3     static synchronized void produce(String s) {
4         while (b!=null) {
5             Main.class.wait();
6         }
7         b = s;
8         System.out.println("Produced " + s);
9         Main.class.notifyAll();
10    }
11    static synchronized void consume() throws InterruptedException {
12        while (b == null) {
13            Main.class.wait();
14        }
15        System.out.println("Consumed " + b);
16        b = null;
17        Main.class.notify();
18    }
19    public static void main(String[] args) {
20        Thread t1 = new Thread(() -> {
21            try {
22                produce("P1");
23                produce("P2");
24                produce("P3");
25            } catch (Exception e) {
26                System.out.println(e.getMessage());
27            }
28        });
29        Thread t2 = new Thread(() -> {
30            try {
31                consume();
32                consume();
33                consume();
34            } catch (Exception e) {
35                System.out.println(e.getMessage());
36            }
37        });
38        t1.start();
39        t2.start();
40        t1.join();
41        t2.join();

```

```
42     }  
43 }
```

Input

stdin input

Output

```
Produced P1  
Consumed P1  
Produced P2  
Consumed P2  
Produced P3  
Consumed P3
```

✓ Submitted

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